2026-11-07 22:53:36 **You**

what eigen vector and value show with visualization and how calculated

ChatGPT gpt-6

In PCA, **eigenvectors show the directions in which the data varies**, while **eigenvalues show how much variance exists in each of those directions**.

Think of it this way:

- Eigenvector = Direction of a road.
- Eigenvalue = How much traffic (variation) flows along that road.

Let's understand this using our previous PCA example.

1. Visual understanding

Our original dataset was:

Observation	X1	X2
1	2	1
2	3	2
3	4	3
4	5	4
5	6	5

After mean-centering:

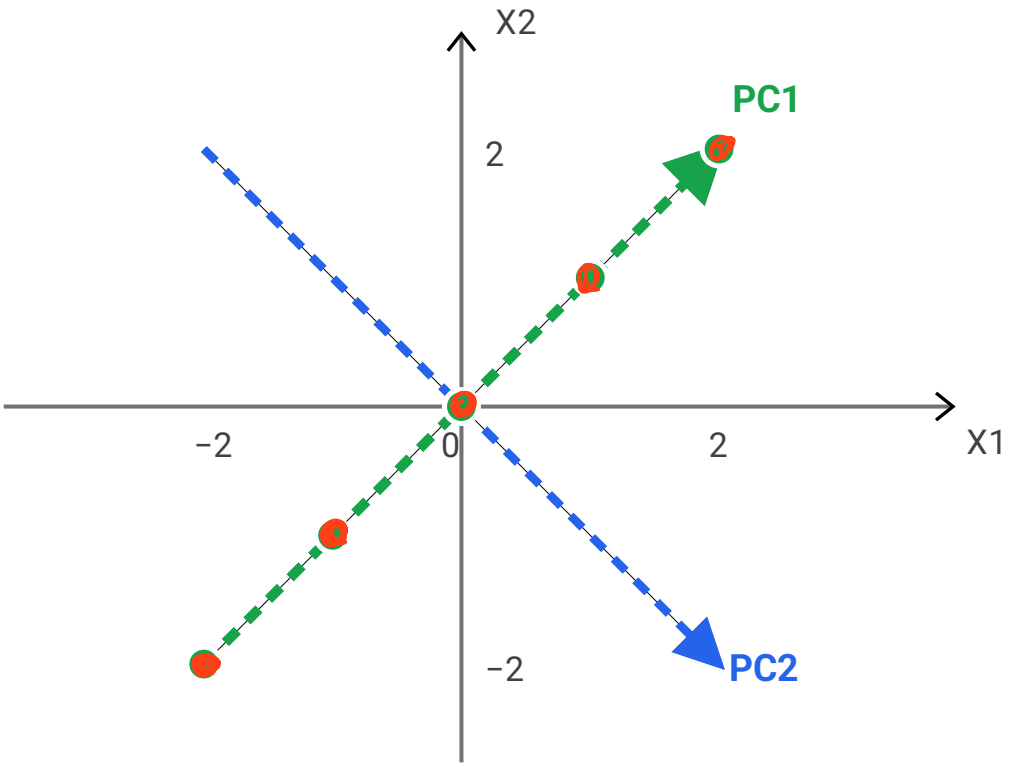
$\bar{x}_1 = 4 \quad \bar{x}_2 = 3$

$x_{1cen} = X1 - \bar{x}_1 \qquad x_{2cen} = X2 - \bar{x}_2$

−2	−2
−1	−1
0	0
1	1
2	2

Notice that the data points follow a diagonal line.

Data distribution and eigenvector directions



Centered data points

PC1: Main direction PC2: Perpendicular direction

Conceptual illustration. The eigenvectors are perpendicular and pass through the data mean (the origin after centering).

What do you observe?

- PC1 follows the direction where the data points spread the most.
- PC2 is perpendicular to PC1, where there is no spread in this particular dataset.

These two directions are eigenvectors.

2. What does an eigenvector show?

An eigenvector represents a direction in the feature space.

For our example, the covariance matrix was:

$$C = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

We calculated two eigenvectors:

$$v_1 = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

Exported with AI Exporter

$$v_2 = \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}$$

Interpretation:

- v_1: Both X1 and X2 contribute positively to PC1.
- v_2: X1 and X2 contribute in opposite directions to PC2.

The values 0.707 are called the eigenvector coefficients or loadings (for normalized eigenvectors).

3. What does an eigenvalue show?

An eigenvalue measures the variance of the data along its corresponding eigenvector.

Our calculated eigenvalues were:

$$\lambda_1 = 4$$

$$\lambda_2 = 0$$

Variance along each direction

PC1 **4**

100% of total variance

PC2 **0**

0% of total variance

Conclusion: PC1 contains all the variance, while PC2 contains none.

That is why PCA keeps PC1 and discards PC2 in this example.

4. How are eigenvalues calculated?

This is the important mathematical part for your exam.

We start with the covariance matrix:

$$C = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

The eigenvalue equation is:

$$\boxed{|C - \lambda I| = 0}$$

Here, I is the identity matrix:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Step 1: Subtract λI from C.

$$\begin{vmatrix} 2 - \lambda & 2 \\ 2 & 2 - \lambda \end{vmatrix} = 0$$

Step 2: Calculate determinant.

Remember:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Therefore:

$$(2 - \lambda)^2 - (2 \times 2) = 0$$

$$(2 - \lambda)^2 - 4 = 0$$

Step 3: Solve the equation.

$$4 - 4\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 4\lambda = 0$$

$$\lambda(\lambda - 4) = 0$$

Therefore:

$$\boxed{\lambda_1 = 4, \quad \lambda_2 = 0}$$

5. How are eigenvectors calculated?

After finding eigenvalues, we calculate their corresponding eigenvectors using:

$$(C - \lambda I)v = 0$$

For λ = 4:

$$\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

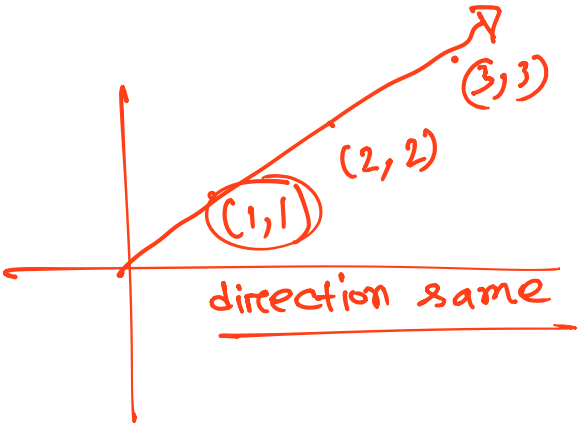
2x2 2x1

2x1

From the first row:

$$-2x + 2y = 0$$

$$x = y$$



Choose $x = 1, y = 1$.

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

So, now the length can be anything so we need to normalize

Normalize it:

$$v_1 = \frac{1}{\sqrt{1^2 + 1^2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\boxed{v_1 = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}}$$

formula

$\bar{v} = \frac{v}{\|v\|}$ → length

$v_1 = \frac{1}{\sqrt{2}} = 0.707$

For λ = 0:

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$2x + 2y = 0$$

$$x = -y$$

Choose $x = 1, y = -1$.

After normalization:

$$\boxed{v_2 = \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}}$$

6. How does PCA choose which eigenvector to keep?

It sorts eigenvalues from largest to smallest.

$$EV_i = \frac{\lambda_i}{\sum \lambda} \times 100$$

Eigenvalue	Eigenvector	Explained variance	Decision
4	(0.707, 0.707)	100%	Keep
0	(0.707, −0.707)	0%	Discard

The eigenvector with the largest eigenvalue becomes the first principal component.

7. Test your understanding

Question: Suppose a PCA calculation gives these eigenvalues:

- PC1 = 8
- PC2 = 2
- PC3 = 0

Which component should we select if we want to retain 80% of the variance?

- (●) Only PC1
- ( ) PC1 and PC2
- ( ) All three components
- ( ) Only PC3

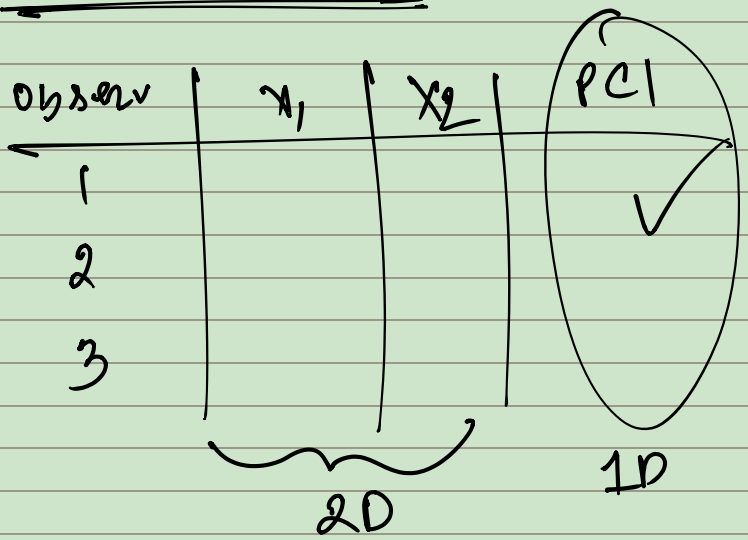
Check answer

The most important thing to remember for your exam:

Mathematical result	What it tells you
Eigenvector	Direction of a principal component
Eigenvalue	Amount of variance in that direction
Largest eigenvalue	Most important principal component
Eigenvector with largest eigenvalue	Direction selected first by PCA

If you understand these two concepts, the mathematical logic behind PCA becomes much easier.

What happened



Before PCA

